Unit 5, Lesson 4: Writing and Solving Real-World Equations

Benchmark

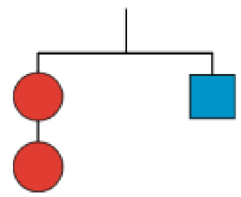
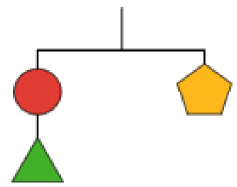
MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

Learning Target

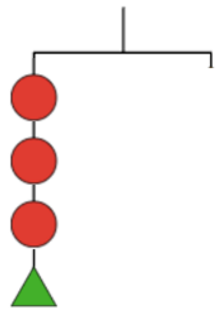
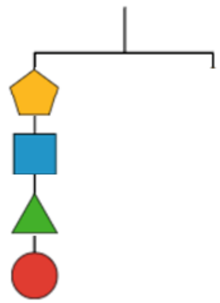
☐ I can write and solve two-step equations given a real-world context.

5.4.1 Warm-Up: Hanger Diagrams

Two hanger diagrams are shown. They are balanced with an equal weight hanging from both sides.

Diagram A	Diagram B
	

1. Use the diagrams above to complete the right side of each diagram so that Diagrams C and D are balanced. You cannot copy the left side of the diagram.

Diagram C	Diagram D
	

2. James drew a pentagon and a square for the right side of Diagram C. Draw pictures or explain how James could conclude that a pentagon and a square are equivalent to the left side of Diagram C.

5.4.2 Guided Instruction: Solving Real-World Problems by Writing and Solving Equations

1. The U.S. Census Bureau recorded a population of approximately 18.8 million people living in Florida in 2010. Since 2010, there has been an average increase of 0.3 million people per year.

   a. What are the quantities that are changing in this scenario?

   b. Define the variables a and p.

   c. What is the parameter?

   d. What is the constant?

   e. Which equation represents the situation?

      ○ $p = 18.8a + 0.3$
      ○ $p = 0.3a + 18.8$
      ○ $a = 18.8p + 0.3$
      ○ $a = 0.3p + 18.8$

   f. Use your equation to determine in what year Florida's population was approximately 21.2 million.

2. Simone purchased 4 skinny vanilla lattes for her and her three friends. A 6% sales tax was applied to the purchase, and she gave the rest of her money as a $\$3.86$ tip for a total of $\$24$.

a. Define the variable in this context.

b. Discuss with your partner what the parameters are in this context.

c. Complete the table to write an equation that represents the context.

Total $ Spent	$=$	**# of Lattes**	**(Latte Price)**	**(1 + Tax Rate)**	$+$	**Tip**
	$=$				$+$	

d. Solve the equation for the unknown.

e. Interpret the solution.

5.4.3 Guided Practice: Solving Real-World Problems by Writing and Solving Equations

1. The popularity of Twitter can be measured using the number of active accounts. In January of 2015, Twitter had approximately 302.5 million active accounts worldwide. Each year, the number of active accounts has increased by about 10.2 million.

a. Define the variables in this context.

b. What is the parameter?

c. What is the constant?

d. Which variable is the output in this context?

e. Write an equation to represent this situation.

f. Write a one-variable equation that can be used to determine the year Twitter will have 425 million active accounts if the trend continues.

g. Solve the equation from Part F.

h. Interpret the solution.

i. Are there any constraints to the given situation?

5.4.4 Try It: Solving Real-World Problems by Writing and Solving Equations

1. Since the fourth quarter of 2014, the flash storage capacity of smartphones has grown by approximately 12 gigabytes a year. The average flash storage in 2014 was 15.9 gigabytes.

   a. Write an equation to represent this situation, defining both variables.

   b. Assuming the trend continues, write a one-variable equation that can be used to find the average flash storage capacity of smartphones in 2025.

   c. Solve your equation from Part B.

   d. Interpret the solution.

   e. Discuss any constraints of the situation with your partner.

5.4.5 Wrap-Up: Ticket Out the Door

1. James is going to play paintball. He is charged $25.95 for equipment rental, plus $6.57 per box of 100 paintballs. If James spent $58.80, how many boxes of paintballs did he purchase?

   a. Write and solve an equation representing this situation.

   b. Interpret the solution.

Unit 5, Lesson 5: Equations and Solutions in the Real World

Benchmarks

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

Learning Targets

- ☐ I can write and solve multi-step equations in a real-world context.
- ☐ I can interpret solutions as viable or non-viable in a given context.